

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Service tool, Part Number 3164088, or equivalent
- Hydraulic Camshaft Bushing Tool Kit, Part Number 3823621, or equivalent
- Valve Camshaft Bushing Replacer Kit, Part Number 3164088, or equivalent
- Receiver spacer, Part Number 3824864, or equivalent

Additional Service Items

- No additional service items required.

Preparatory Steps

⚠ WARNING ⚠

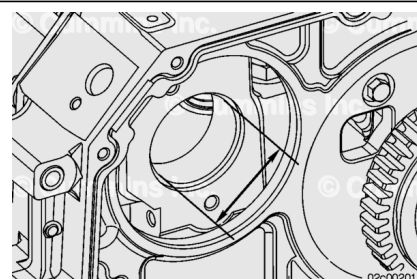
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the valve camshaft. Refer to Procedure 002-024 in Section 2.
(/qs3/pubsys2/xml/en/procedures/377/377-002-024.html)

Initial Check

Measure the inner diameter of the installed bushings.

Valve Camshaft Bushings Inside Diameter - Installed		
mm		in
65.059	MIN	2.561
65.123	MAX	2.564

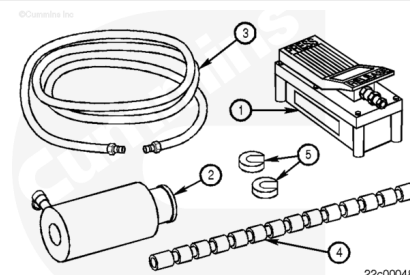


If the bushings are **not** within specifications, the bushings **must** be replaced.

Remove

Use service tool, Part Number 3164088, or equivalent, to remove the valve camshaft bushings.

The hydraulic camshaft bushing tool kit, Part Number 3823621, or equivalent, consists of:



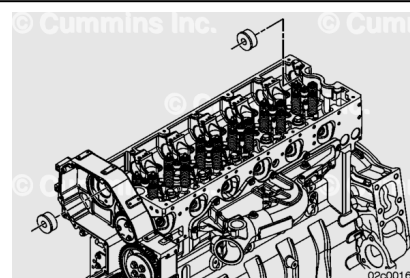
Item	Part Number	Description	Qty.
1	3823622	Pump, air hydraulic	1
2	4918216	Center hole ram	1
3	3823624	Hydraulic hose	1
4	3823625	Puller shaft	1
5	3823626	Stop collar	2

- For inner camshaft bushing removal, follow Steps 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10.
- For rear camshaft bushing removal, follow Steps 1, 2, 3, 4, 5, 6, 11, 12, and 13.
- For front camshaft bushing removal, follow Steps 1, 2, 3, 4, 5, 6, 14, 15, and 16.

Step 1:

⚠ CAUTION ⚠

Make sure to select the proper-size centering guides. This kit contains different-size centering guides for journals both with and without bushings installed. Selection of incorrect centering guides can result in damage to the journal and/or guides.

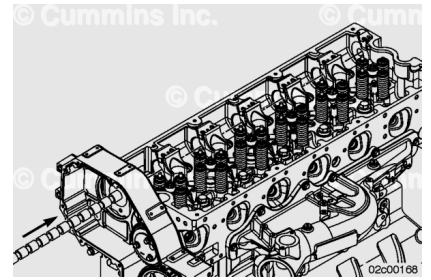


Install the centering guides in two of the camshaft bore ends.

Note : The centering guide can be installed in any of the holes, but keeping the wider distance is preferred.

Step 2:

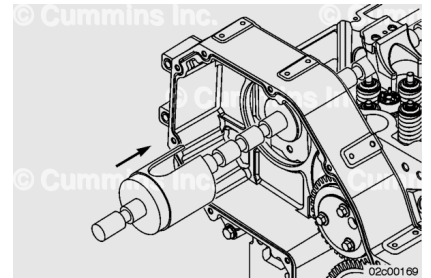
Install the puller shaft through both of the installed centering guides.



Step 3:

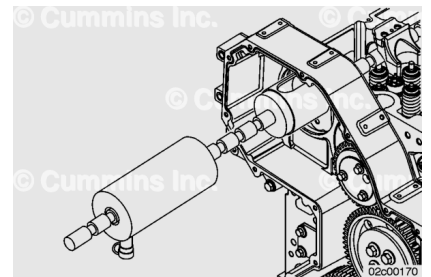
Position the receiver spacer on the puller shaft with the open end facing the cylinder head.

Note : A receiver spacer has **not** been included in the valve camshaft bushing replacer kit, Part Number 3164088. Use receiver spacer, Part Number 3824864, or an appropriate size spacer from another kit.



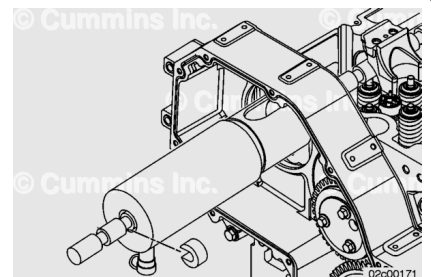
Step 4:

Slide the center hole ram onto the puller shaft until it touches the receiver spacer.



Step 5:

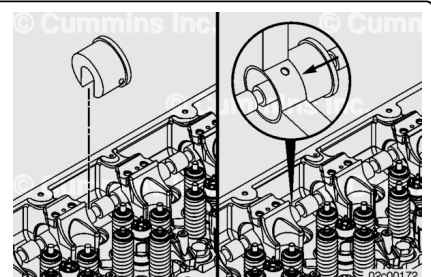
Install the stop collar in the groove of the puller shaft closest to the center hole ram.



Step 6:

⚠ CAUTION ⚠

Make sure to install the proper-size bushing driver. Selection of incorrect bushing driver can result in damage to the camshaft bushing, journal cylinder head bore, and tool equipment.

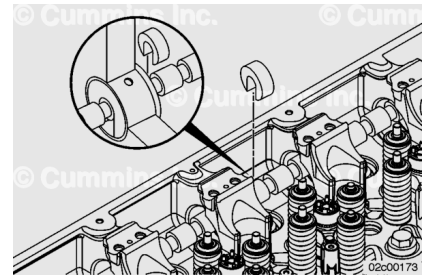


Position the bushing driver on the puller shaft next to the bushing that is being removed.

Slide the bushing driver into the camshaft bushing until it stops against the flange of the bushing driver.

Step 7: (for inner bushings **only)**

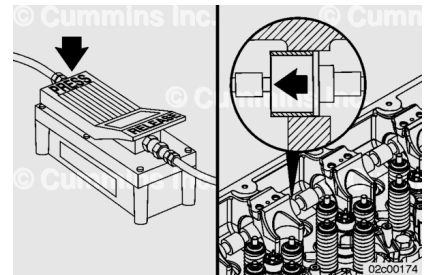
Install the stop collar into the groove on the puller shaft closest to the end of the bushing driver.



Step 8: (for inner bushings **only)**

⚠ WARNING ⚠

Keep hands clear of the journal area when operating the pedal of the hydraulic pump. Serious injury can result if hands are caught between journal and bushing area when operating the hydraulic pump pedal.

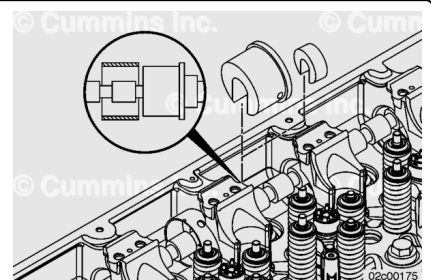


Push down on the PRESS side of the air/hydraulic pump pedal and hold until the camshaft bushing is out of the bore.

Step 9: (for inner bushings **only)**

Remove the camshaft bushing from the bushing driver.

Remove the bushing driver and stop collar from the puller shaft.

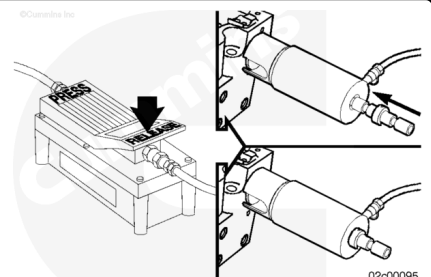


Step 10: (for inner bushings **only)**

Push down on the RELEASE side of the air/hydraulic pump pedal. This will release the pressure.

Push the puller shaft forward until the stop collar is touching against the center hole ram.

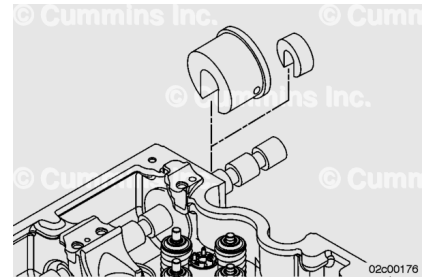
Remove the service tool from the engine.



Step 11: (for rear bushing **only)**

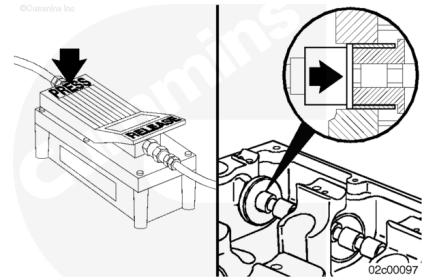
Install the bushing driver onto the puller shaft behind the outer camshaft bushing.

Install the stop collar into the groove on the puller shaft closest to the end of the bushing driver.



Step 12: (for rear bushing *only*)

Push down on the PRESS side of the air/hydraulic pump pedal and hold until the rear camshaft bushing and centering guide are out of the bore.

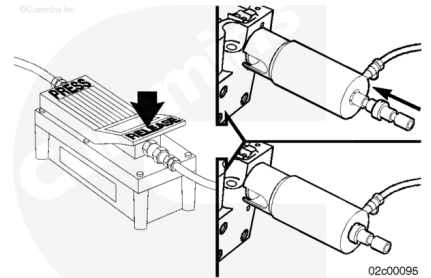


Step 13: (for rear bushing *only*)

Push down on the RELEASE side of the air/hydraulic pump pedal. This will release the pressure.

Push the puller shaft forward until the stop collar is touching against the center hole ram.

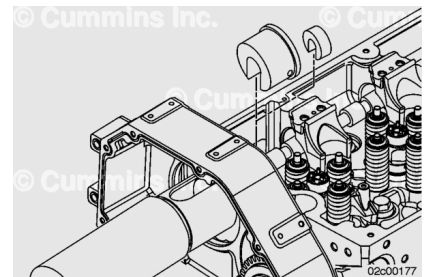
Remove the service tool from the engine.



Step 14: (for front bushing *only*)

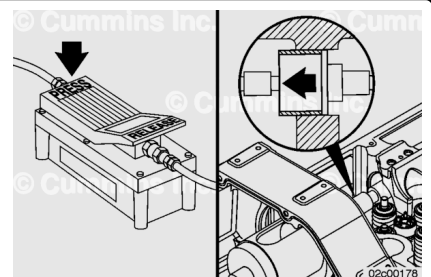
Install the bushing driver on the puller shaft so that the larger end of the bushing driver is facing the front camshaft bushing.

Install the stop collar into the groove on the puller shaft closest to the end of the bushing driver.



Step 15: (for front bushing *only*)

Push down on the PRESS side of the air/hydraulic pump pedal and hold until the camshaft bushing and centering guide are out of the bore.

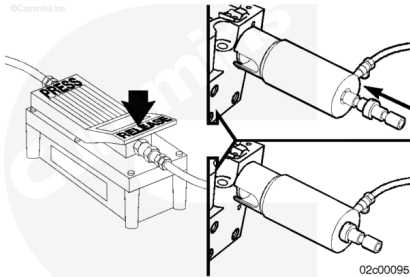


Step 16: (for front bushing *only*)

Push down on the RELEASE side of the air/hydraulic pump pedal. This will release the pressure.

Push the puller shaft forward until the stop collar is touching against the center hole ram.

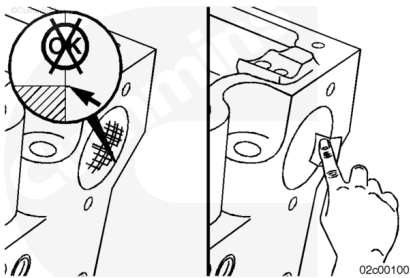
Remove the service tool from the engine.



02c00095

Inspect for Reuse

Clean and inspect the camshaft bores in the cylinder head for reuse. Use an emery cloth to remove any burrs and sharp edges from the bore.

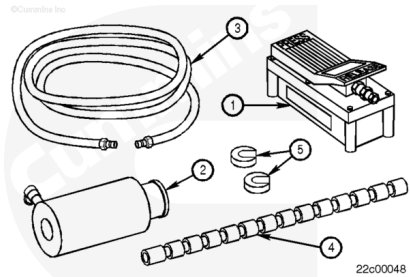


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Install

Use service tool, Part Number 3164088, for valve camshaft bushing installation.

The hydraulic camshaft bushing tool kit, Part Number 3823621, or equivalent, consists of:



22c00048

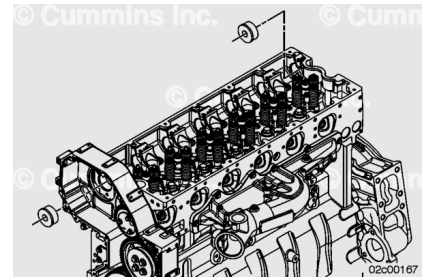
Item	Part Number	Description	Qty.
1	4918216	Pump, air hydraulic	1
2	3823623	Center hole ram	1
3	3823624	Hydraulic hose	1
4	3823625	Puller shaft	1
5	3823626	Stop collar	2

- For inner camshaft bushing installation, follow Steps 1, 2, 5, 6, 7, 8, 9, 10, and 11.
- For rear camshaft bushing installation, follow Steps 1, 3, 5, 6, 7, 12, 13, 14, and 15.
- For front camshaft bushing installation, follow Steps 1, 4, 5, 6, 7, 16, 17, 18, and 19.

Step 1:

⚠ CAUTION ⚠

Make sure to select the proper-size centering guides. This kit contains different-size centering guides for journals both with and without bushings installed. Selection of incorrect centering guides can result in damage to the journal and/or guides.



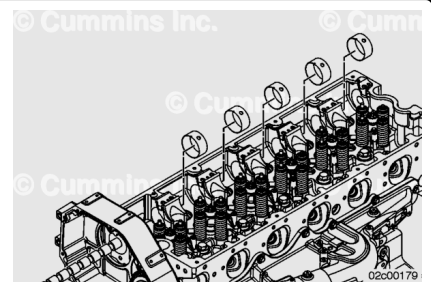
Install the centering guides into both camshaft bore ends.

Step 2: (for rear bushing **only**)

Slide the puller shaft through the centering guide and into the number 2 journal. Place a camshaft bushing, if desired, over the end of the puller shaft.

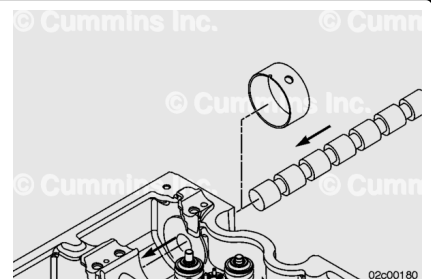
Move the puller shaft through the number 3 journal and do the same as above.

Continue through each inner journal until the puller shaft is completely through the opposite-end centering guide.



Step 3: (for rear bushing **only**)

Install the puller shaft through the two installed centering guides.

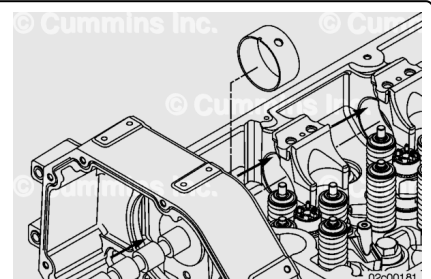


Step 4: (for front bushing **only**)

Slide the puller shaft through the centering guide and into the number 1 journal.

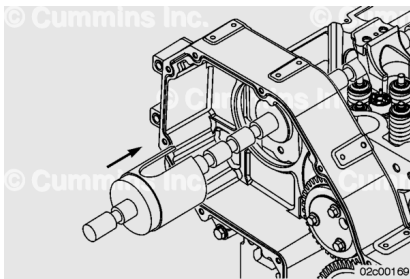
Place a camshaft bushing over the end of the puller shaft.

Continue to slide the puller shaft through the block and opposite-end centering guide.



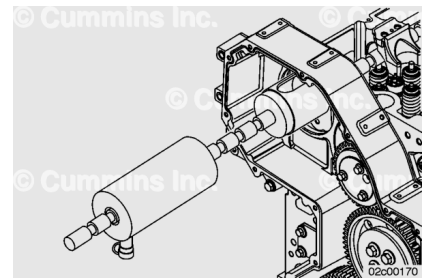
Step 5:

Position the receiver spacer on the puller shaft with the open end facing the cylinder head.



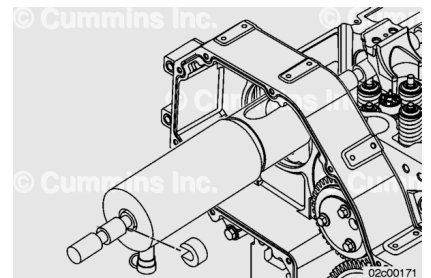
Step 6:

Slide the center hole ram onto the puller shaft until it touches the receiver spacer.



Step 7:

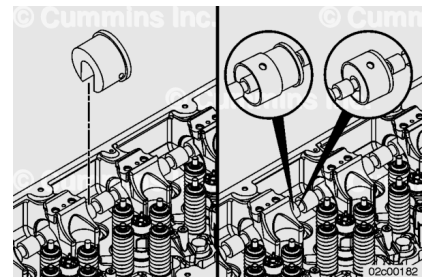
Install the stop collar in the groove of the puller shaft closest to the center hole ram.



Step 8: (for inner bushings **only**)

⚠ CAUTION ⚠

Make sure to install the proper-size bushing driver. Selection of incorrect bushing driver can result in damage to the camshaft bushing, journal, and tool equipment.



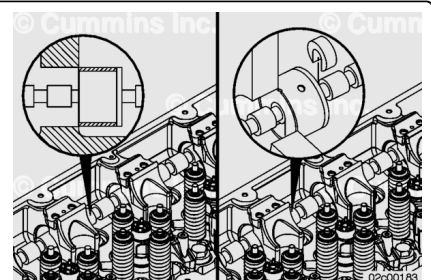
Position the proper bushing driver on the puller shaft next to the bushing being installed.

Slide the camshaft bushing over the bushing driver.

Step 9: (for inner bushings **only**)

⚠ CAUTION ⚠

It is very important that the oil ports on the camshaft bushings being installed be aligned with the oil ports in the journal. Misalignment of the camshaft bushing oil port to the journal oil port can result in damage to the engine.



Push the bushing driver over until the camshaft bushing just touches the cylinder head.

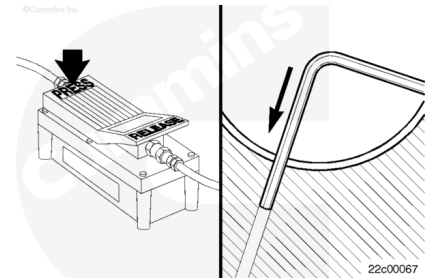
Install the stop collar into the groove on the puller shaft closest to the end of the bushing driver.

Rotate the camshaft bushing until the oil port is aligned with the oil port of the bore.

Step 10: (for inner bushings **only**)

⚠ WARNING ⚠

Keep hands clear of the journal area when operating the pedal of the hydraulic pump. Serious injury can result if hands are caught between journal and bushing area when operating the hydraulic pump pedal.



Push down on the PRESS side of the air/hydraulic pump pedal to pull the camshaft bushing into the cylinder head. As the camshaft bushing oil port comes close to the bore oil port, let up on the pedal.

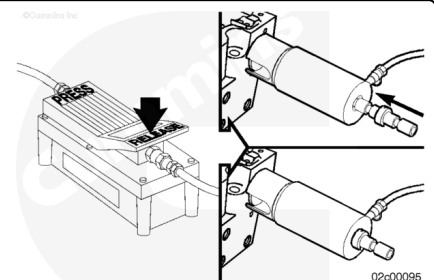
Lightly tapping the PRESS side of the air/hydraulic pump pedal will provide control to stop the camshaft bushing in the desired position.

Step 11: (for inner bushings **only**)

Push down on the RELEASE side of the air/hydraulic pump pedal. This will release the pressure.

Push the puller shaft forward until the stop collar is touching against the center hole ram.

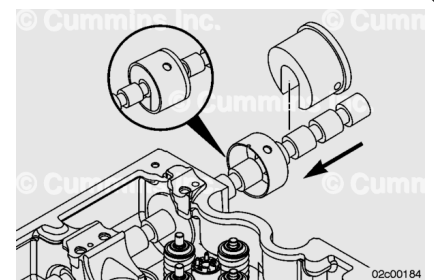
Remove the service tool from the engine.



Step 12: (for rear bushing **only**)

⚠ CAUTION ⚠

Make sure to install the proper-size bushing driver. Selection of incorrect bushing driver can result in damage to the camshaft bushing, journal, and tool equipment.



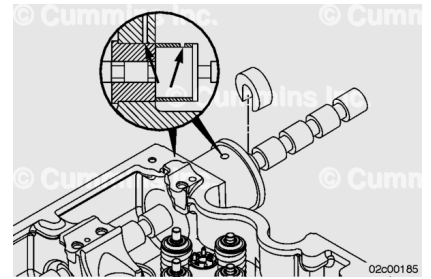
Place the camshaft bushing over the end of the puller shaft.

Position the bushing driver on the puller shaft and slide the camshaft bushing onto the bushing driver.

Step 13: (for rear bushing **only**)

⚠ CAUTION ⚠

It is very important that the oil ports on the camshaft bushings being installed be aligned with the oil ports in the journal. Misalignment of camshaft bushing oil port to journal oil port can result in damage to the engine.



Push the bushing driver over until the camshaft bushing just touches the cylinder head.

Install the stop collar into the groove on the puller shaft closest to the end of the bushing driver.

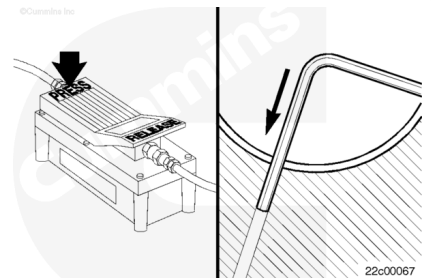
Rotate the camshaft bushing until the oil port is aligned with the oil port of the bore.

Step 14: (for front bushing **only**)

For front and rear bushing installation, move the centering guides to journals 2 and 6.

Push down on the PRESS side of the air/hydraulic pump pedal to pull the camshaft bushing into the cylinder head. As the camshaft bushing oil port comes close to the bore oil port, let up on the pedal.

Lightly tapping the PRESS side of the air/hydraulic pump pedal will provide control to stop the camshaft bushing in the desired position.

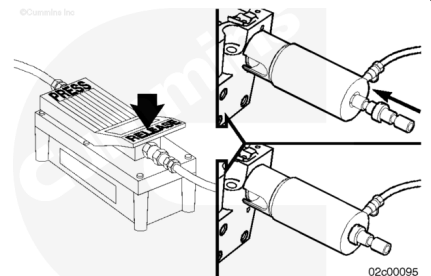


Step 15: (for rear bushing **only**)

Push down on the RELEASE side of the air/hydraulic pump pedal. This will release the pressure.

Push the puller shaft forward until the stop collar is touching against the center hole ram.

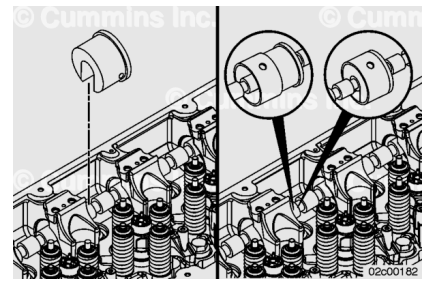
Remove the service tool from the engine.



Step 16: (for front bushing **only**)

Position the proper bushing driver on the puller shaft.

Slide the camshaft bushing over the bushing driver.

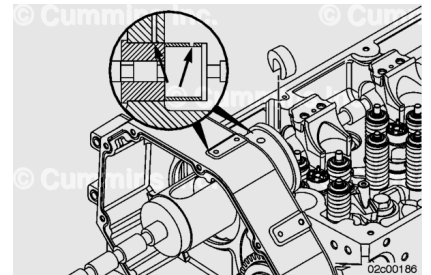


Step 17: (for front bushing **only**)

Push the bushing driver over until the camshaft bushing just touches the cylinder head.

Install the stop collar into the groove on the puller shaft closest to the end of the bushing driver.

Rotate the camshaft bushing until the oil port is aligned with the oil port of the bore.

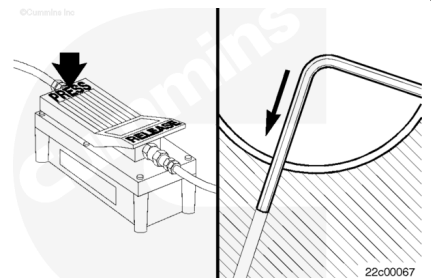


Step 18: (for front bushing **only**)

Note : The centering guide will be forced into the receiver spacer when the bushing and driver are pulled into the bore.

Push down on the PRESS side of the air/hydraulic pump pedal to pull the camshaft bushing into the cylinder head. As the camshaft bushing oil port comes close to the bore oil port, let up on the pedal.

Lightly tapping the PRESS side of the air/hydraulic pump pedal will provide control to stop the camshaft bushing in the desired position.

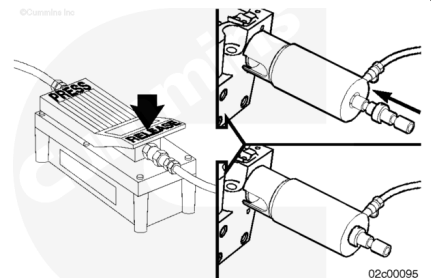


Step 19: (for front bushing **only**)

Push down on the RELEASE side of the air/hydraulic pump pedal. This will release the pressure.

Push the puller shaft forward until the stop collar is touching against the center hole ram.

Remove the service tool from the engine.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the valve camshaft. Refer to Procedure 002-024 in Section 2.
(/qs3/pubsys2/xml/en/procedures/377/377-002-024.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for leaks.